

CMS Project Planning

Project Goal

Build a personal CMS platform where:

- One administrator manages content
- Public users can browse content without login
- Three categories:
 - Experience
 - Learning
 - Lifestyle

Objectives

- Learn software engineering practices
- Learn full-stack development
- Build a maintainable portfolio project
- Integrate third-party services to extend application functionality

Product Backlog (User Stories)

US_ID	As a/an	I want to ...	So that ...	Priority	Story Point
US001	Admin	Create content	I can publish content on the website	High	5
US002	Admin	View content list	I can manage all content records efficiently	High	3
US003	Admin	View content detail	I can select a content item and view its complete article	High	3
US004	Admin	Edit content	I can update existing content when necessary	High	3
US005	Admin	Delete content	Outdated content can be removed from active listings	Medium	2
US006	Admin	Restore deleted content	I can recover content that was removed accidentally	Medium	2
US007	User	Browse published content	I can discover available content	High	3

US008	User	View content detail	I can select a published content item and view its complete article	High	3
US009	User/Admin	Search content by keyword	I can find relevant content quickly	Medium	2
US010	User/Admin	Filter content by category	I can browse content more efficiently	Medium	2
US011	Admin	Filter content by status	I can locate content based on its publication status and visibility status	Medium	2
US012	Admin	Log in securely and access protected pages	Unauthorized users cannot access admin features	High	2
US013	Admin	Preview content before publishing	I can verify layout and formatting	Medium	1
US014	User	View website statistics	I can understand the website's growth and activity	Medium	2
US015	Admin	View visitor analytics dashboard	I can monitor website performance and visitor engagement	Medium	2
US016	Admin	Manage related content and external references	I can enhance the reader's experience by providing relevant recommendations and useful resources	Medium	3
US017	User/Admin	View related content and external references	I can discover other relevant content and access additional resources	Medium	2

Functionalities (Epics)

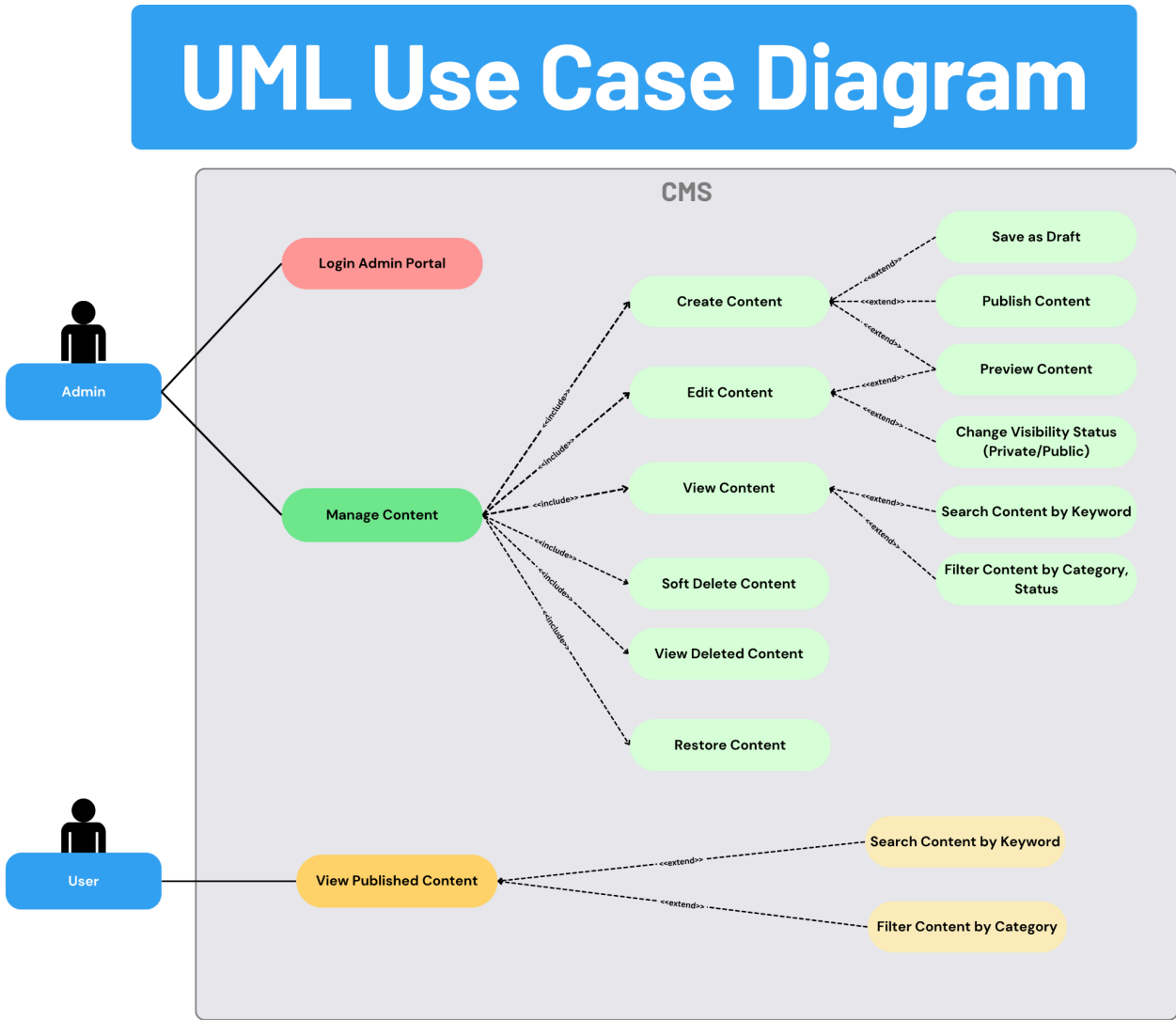
ID	Main Function	Sub-Function Description	Priority	Effort pts
1	Content Management	Manage the creation, viewing, editing, publishing, and lifecycle of website content	High	
1.1		Create Content	High	5
1.1.1		Select Category	High	1

1.1.2		Enter Content Information	High	1
1.1.3		Manage Cover Image (Default or Custom)	High	2
1.1.4		Manage Content Sections	High	2
1.1.5		Save as Draft	High	2
1.1.6		Preview Content	Medium	1
1.1.7		Publish Content	High	2
1.1.8		Manage Related Content	Medium	2
1.1.9		Add External References	Medium	1
1.2		View and Manage Content List	High	3
1.2.1		Search by Keyword	High	1
1.2.2		Filter by Category	High	1
1.2.3		Filter by Content Status	High	1
1.2.4		Filter by Visibility Status	High	1
1.2.5		Pagination	High	1
1.2.6		View Content Detail	High	2
1.2.6.1		View Related Content and External References	Medium	1
1.3		Edit Content	High	5
1.3.1		Modify Content Information	High	1
1.3.2		Manage Cover Image (Default or Custom)	High	2
1.3.3		Manage Content Sections	High	2
1.3.4		Save as Draft	High	2
1.3.5		Preview Content	Medium	1
1.3.6		Publish (Draft Content)	High	2
1.3.7		Save Changes (Published Content)	High	2
1.3.8		Modify Related Content	Medium	2
1.3.9		Modify External References	Medium	1

1.4		Soft Delete Content	Medium	2
1.5		Restore Content	Medium	2
1.5.1		Select Visibility Status (Public, Private)	Medium	1
1.5.2		Restore Soft-Deleted Content	Medium	1
2	Public Content Viewing	Allow visitors to browse and read published content	High	
2.1		Browse Published Content	High	3
2.1.1		Search by Keyword	High	1
2.1.2		Filter by Category	High	1
2.1.3		Pagination	High	1
2.2		View Content Detail	High	2
2.2.1		View Related Content and External References	Medium	1
3	Authentication & Administration	Secure administrative access to the CMS	High	
3.1		Admin Login	High	2
3.1.1		Authenticate Credentials	High	1
3.1.2		Access Protected Pages	High	1
3.1.3		Logout	High	1
4	Website Analytics	Provide website statistics and visitor insights through analytics integration	Medium	
4.1		Public Website Statistics	Medium	2
4.1.1		Display Total Visitors and Page Views	Medium	1
4.1.2		Display Website Growth Metrics	Medium	1
4.2		Admin Analytics Dashboard	Medium	3
4.2.1		View Visitor Trends	Medium	1
4.2.2		View Popular Content	Medium	1
4.2.3		View Category Distribution	Medium	1

4.2.4		View Traffic Sources	Medium	1
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Use Case Diagram



Use Case Specification

ID	Use Case Name
CM-1	Create Content
CM-2	View Content
CM-3	Edit Content

ID	Use Case Name
CM-4	Soft Delete Content
CM-5	Restore Content
AU-1	Admin Login
PV-1	View Published Content

Wireframes

1. Admin Manage Content Page V1

WAI - ZXT

Home | **Manage Content** | Public Website

Content Management

> Create Content

< Search by keyword

Filter by Category V

Filter by Content Status V

Filter by Visibility Status V

ID	Content Title	Category	Content Status	Visibility Status	Actions
1	Title 1	Experience	Published	Public	View Detail Edit Delete
2	Title 2	Learning	Draft	-	Edit Delete
3	Title 3	Lifestyle	Soft-Deleted	-	View Detail Restore
4	Title 4	Learning	Published	Private	View Detail Edit Delete

<<

1 of 2

10 Items per page V

>>

2. Create Content Page V1

Content Creation

Content Overview

Category: Experience V

Title: Title 1

Description: Description 1...

Cover Photo:

Select Default Image

Default 1 Default 2 Default 3 

OR

Upload Custom Image

Choose file

Selected Image

Default 3

Section 1 Remove

Section Title: Section Title 1

Section Description: Section Description 1...

Section Photo:

Select Default Image

Default 1 Default 2 Default 3

OR

Upload Custom Image

Choose file

Selected Image

Custom

+ Add Section

Save as Draft Preview Cancel Publish

3. Edit Content Page V1
 - a. Edit Draft Content

Edit Content

Content Overview

Category: Experience V

Content Status: Published V

Viability Status: Public V

Title: Title 1

Description: Description 1...

Cover Photo:

Select Default Image

Default 1 Default 2 Default 3 

OR

Upload Custom Image

Choose file

Selected Image

Default 3

Section 1 Remove

Section Title: Section Title 1

Section Description: Section Description 1...

Section Photo:

Select Default Image

Default 1 Default 2 Default 3

OR

Upload Custom Image

Choose file

Selected Image

Custom

+ Add Section

Save as Draft Preview Cancel Publish

- b. Edit Published Content

Edit Content

Content Overview

Category:

Experience

V

Content Status:

Published

V

Visibility Status:

Public

V

Title:

Title 1

Description:

Description 1....

Cover Photo:

Select Default Image

Default 1

Default 2

Default 3

OR

Upload Custom Image

Choose file

Selected Image

Default 3

Section 1

Remove

Section Title:

Section Title 1

Section Description:

Section Description 1....

Section Photo:

Select Default Image

Default 1

Default 2

Default 3

OR

Upload Custom Image

Choose file

Selected Image

Custom

+ Add Section

Preview

Cancel

Publish

c. **Updated:** Image Edit (same as Section Photo)

Cover Photo:

Current Image [clickable]

Default 3

Select Default Image

Default 1

Default 2

Default 3

OR

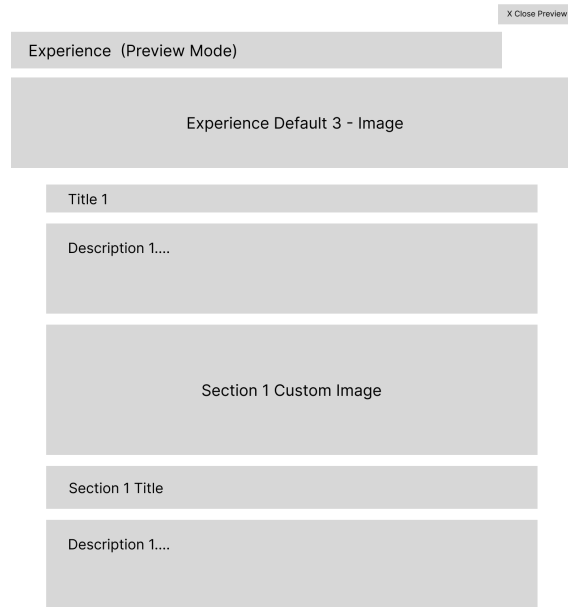
Upload Custom Image

Choose file

New Selected Image

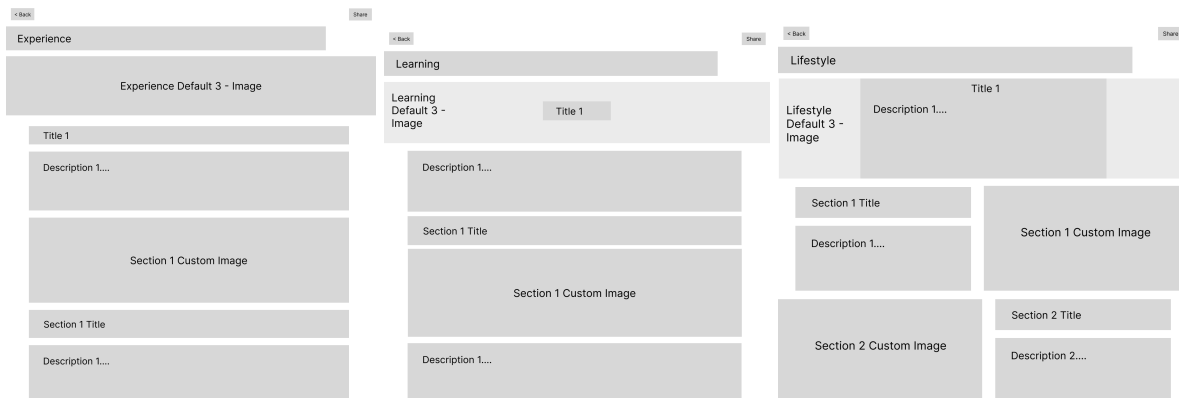
Default 1

4. Content Preview Page V1

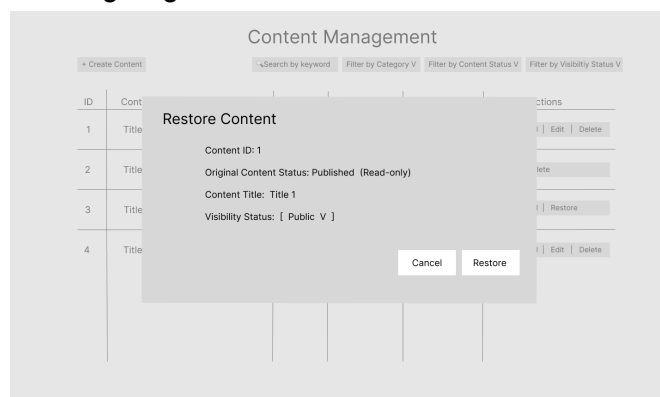


5. Admin Content Detail Page V1

6. Public Content Detail Page V1



7. Restore Content Dialog Page V1



8.

a. Public Home Page V1

WAI - ZXT

Welcome to my CMS Portfolio

Explore my experiences, learning journey and lifestyle articles.

Browse Content

Total Published Content

5

Experience	Learning	Lifestyle
3	1	1

Latest Content

	Title	Category
	Description	

Published Content

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

1 of 6

1 of 6

1 of 6

b. Public Home Page V2

WAI - ZXT

Welcome to My Personal CMS

Explore my experiences, learning journey and lifestyle articles.

Browse Content

Display 9 Default Images Here.

Growing Community

16402

Total Readers

65931

Content Views

42

Published Content

3

Categories

V1 View

600

500

400

300

200

100

0

Jan

Feb

Mar

Apr

May

Jun

Latest Content

	Title	Category
	Description	

Published Content

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

Cat

Title

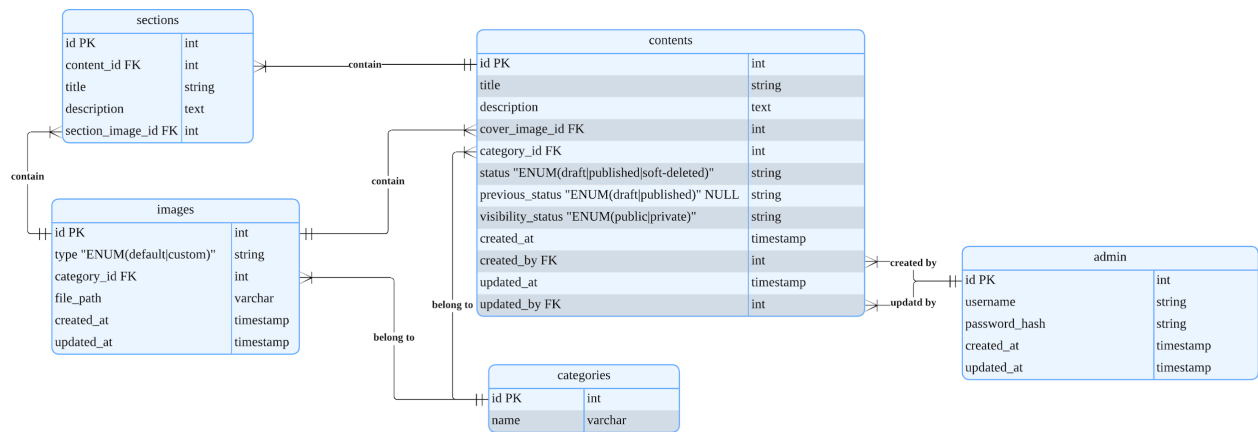
1 of 6

1 of 6

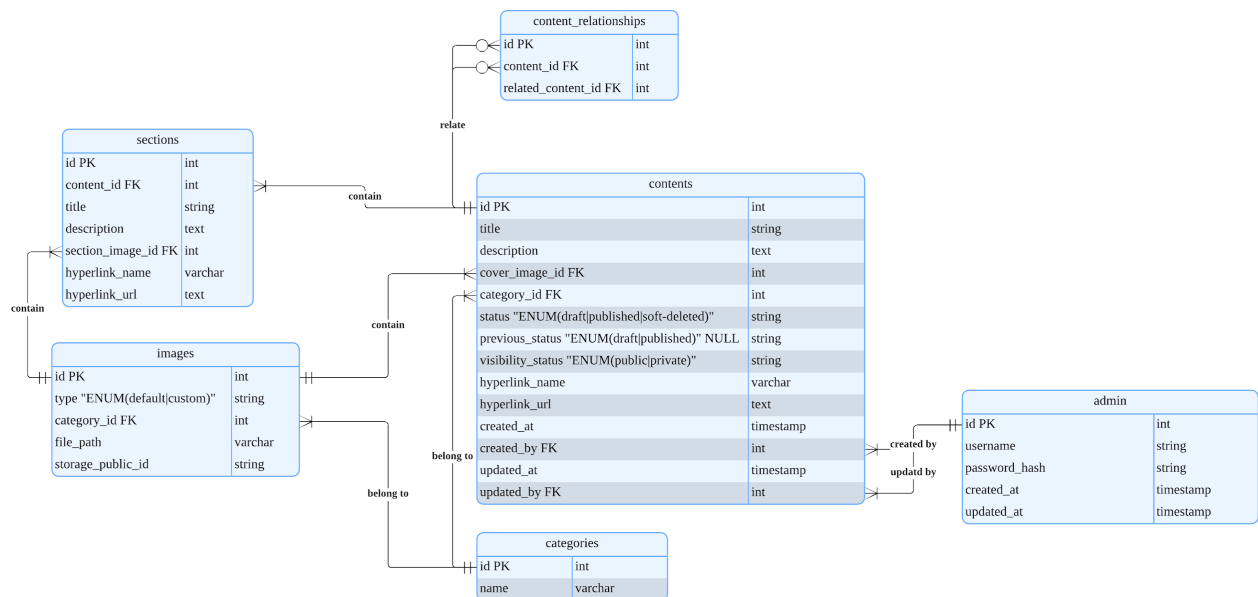
1 of 6

Entity Relationship Diagram

V1



V2



SQL Table Design (Updated)

```
CREATE TABLE categories (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(50) NOT NULL UNIQUE
);
INSERT INTO categories (name)
VALUES
('Experience'),
('Learning'),
('Lifestyle');
```

```
CREATE TABLE admins (
  id INT AUTO_INCREMENT PRIMARY KEY,
  username VARCHAR(100) NOT NULL UNIQUE,
```

```
password_hash VARCHAR(255) NOT NULL,  
  
created_at TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP  
ON UPDATE CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE images (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  
  type ENUM(  
    'default',  
    'custom'  
  ) NOT NULL,  
  
  category_id INT NOT NULL,  
  
  file_path VARCHAR(255) NOT NULL,  
  
  storage_public_id VARCHAR(255) NULL,  
  
  CONSTRAINT fk_images_category  
    FOREIGN KEY (category_id)  
    REFERENCES categories(id)  
);
```

```
CREATE TABLE contents (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  
  title VARCHAR(255) NOT NULL,  
  
  description TEXT NOT NULL,  
  
  cover_image_id INT NOT NULL,  
  
  category_id INT NOT NULL,  
  
  status ENUM(  
    'draft',  
    'published',  
    'soft-deleted'  
  ) NOT NULL,  
  
  previous_status ENUM(  
    'draft',  
    'published'  
  ) NULL,  
  
  visibility_status ENUM(  

```

```

        'public',
        'private'
    ) NOT NULL,

    hyperlink_name VARCHAR(255) NULL,
    hyperlink_url TEXT NULL,

    created_at TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP,

    created_by INT NOT NULL,

    updated_at TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP
        ON UPDATE CURRENT_TIMESTAMP,

    updated_by INT NOT NULL,

    CONSTRAINT fk_contents_cover_image
        FOREIGN KEY (cover_image_id)
        REFERENCES images(id),

    CONSTRAINT fk_contents_category
        FOREIGN KEY (category_id)
        REFERENCES categories(id),

    CONSTRAINT fk_contents_created_by
        FOREIGN KEY (created_by)
        REFERENCES admins(id),

    CONSTRAINT fk_contents_updated_by
        FOREIGN KEY (updated_by)
        REFERENCES admins(id)
);

CREATE TABLE sections (
    id INT AUTO_INCREMENT PRIMARY KEY,

    content_id INT NOT NULL,

    title VARCHAR(255) NOT NULL,

    description TEXT NOT NULL,

    section_image_id INT NOT NULL,

    hyperlink_name VARCHAR(255) NULL,
    hyperlink_url TEXT NULL,

    CONSTRAINT fk_sections_content
        FOREIGN KEY (content_id)

```

```

REFERENCES contents(id),

CONSTRAINT fk_sections_image
FOREIGN KEY (section_image_id)
REFERENCES images(id)
);

CREATE TABLE content_relationships (
  id INT AUTO_INCREMENT PRIMARY KEY,
  content_id INT NOT NULL,
  related_content_id INT NOT NULL,

  CONSTRAINT fk_relationship_content
  FOREIGN KEY (content_id)
  REFERENCES contents(id),
  CONSTRAINT fk_relationship_related_content
  FOREIGN KEY (related_content_id)
  REFERENCES contents(id),

  CONSTRAINT uq_content_relationship
  UNIQUE (content_id, related_content_id),

  CONSTRAINT chk_not_self_related
  CHECK (content_id <> related_content_id)
);

```

Sprint Backlog

Sprint 1	Foundation		
Sprint Goal	Establish the project foundation and implement admin authentication, content creation and content management functionalities.		
	User Story ID	Task	Progress
		Database Setup	Done
		Default and Custom Image Management Setup	Done
	US012	Admin Login and Access	Done
	US001	Create Content	Done
	US002	View Content List	Done
	US009	Search Content by Keyword	Done
	US010	Filter Content by Category	Done

	US011	Filter Content by Content Status and Visibility Status	Done
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Sprint 2	Content Management		
Sprint Goal	Enable administrators to manage existing content through viewing, editing, previewing, deleting and restoring content.		
	User Story ID	Task	Progress
	US003	View Content Detail	Done
	US004	Edit Content	Done
	US005	Soft Delete Content	Done
	US006	Restore Content	Done
	US013	Preview Content	Done

Sprint 3	Public Website		
Sprint Goal	Enable public users to browse published content and view website statistics.		
	User Story ID	Task	Progress
	US007	Browse Published Content	Done
	US008	View Content Detail	Done
	US009	Search Content by Keyword	Done
	US010	Filter Content by Category	Done
	US014	View Website Statistics	Done
		System Testing 1	Done

Sprint 4	Enhancement & Release		
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Sprint Goal	Enhance the CMS with analytics, content recommendations, external references, and complete final testing, deployment, and project documentation.		
	User Story ID	Task	Progress
	US015	Admin Analytics Dashboard	Done
	US016	Manage Related Content and External References	Done
	US017	View Related Content and External References	Done
		Fix: Changing a content photo will delete old and create new photo in Cloudinary.	Done
		System Testing 2	Done
		Deployment	Done
		Documentation	Done